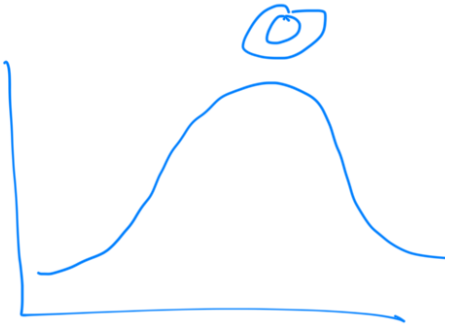
* Revision:

① Covariance: $\frac{\sum (x-\mu_x)(y-\mu_y)}{n}$



+ve: ↑ ↓

-ve: ↑↓ ↓↑

0: the lines r

Scale magnitude

|| Strength ||

② Correlation: $\frac{cov(x,y)}{\sigma_x * \sigma_y}$

Scale

+1 & -1

+1 = +ve

0 → No lines

-1 = -

Heat map

③ Can we find correlation between numeric column & categorical column ?

↳ → A → 1, B → 2, C → 3

IQ

CN

OD →

A	B	C	D
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

* Today's class:

* **Random Exp:**

- we know the all possible outcome but we can't predict it.

Ex: Tossing a coin



* **Random variable:**

Function that asigns a real number to each outcome in the sample space of Random Exp.

Ex ① Toss a coin :

S.S { Head → 1, Tail → 0 } 3 Real values

$X : S \to R$

$P(X=1) = P(H)$

$P(X=0) = P(Tail)$

② Box :

6 Red balls

3 blue balls

2 black balls

5 green balls

If we select one ball Randomly

S { Red → R { 1 }, blue → 2, black → 3, green → 4 }

$X : S \rightarrow R$

$P(X=1) = P(Red) = 6/16$

$P(X=2) = P(blue) =$

* Two types of Random variable :

① Discrete Random variable :

- Takes finite values. S.S = Countable

Ex: same as above

→ (H T) 3 Random variable have only two values 1 & 0.

box 3 R.v. will take only 4 values.

$P(X=1)$
$P(X=2)$
$P(X=3)$
$P(X=4)$

} Fix number.

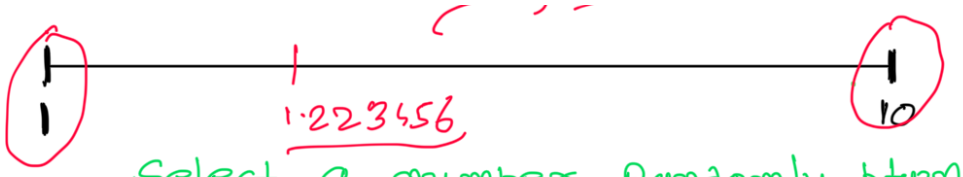
(S) → 1

② Continuous Random variable :

- Variabe that can take any values with in a given interval. (∞)

Ex:

→ Random



Select a number Randomly btwn 1 to 10 : ∞

How many numbers are possible ?

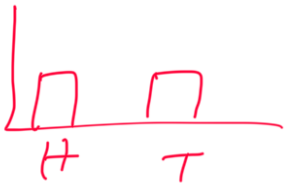
* Probability Distribution:

List of all possible outcome along with their pro. values.

Eg:

H	1/2
T	1/2

pro. dis.



Die	
1	1/6
2	1/6
3	1/6
4	1/6
5	1/6
6	1/6

Pro. dist.

→ we can draw a graph also:



Pro. dis. Die

– is it always possible to list down outcome of R.V. & its pro. ?

→ D.R.V. ↓ C.R.V

* Prob. dist. Function:

- Math. Function that describes the prob. of obtaining diff. values of a Random variable.

R.E → R.V. (S-S → Real No.)

Prob. dis

P.D.F (DC) → P(X = x)

① Prob. Mass Function:

- describe the P.D. of discrete R.V.
- PMF assigns prob. to each possible value of the R.V.

Example 1: Tossing a fair coin

H : 1 → 1
T : 0 → 0

Suppose we toss a coin once.

Let X be the number of heads obtained. X : no. of

Possible values: $X = 0$ or $X = 1$.

Number of heads (X)	Probability $P(X = x)$
0 (Tails)	0.5
1 (Heads)	0.5
Total	**1.0**

The PMF is: 0/1

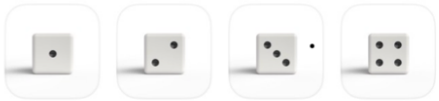
$$P(X = x) = \begin{cases} 0.5 & x = 0 \\ 0.5 & x = 1 \end{cases}$$

0 otherwise

Why is this a PMF? Every probability is between 0 and 1, and the sum of all probabilities is 1.

Example 2: Rolling a fair six-sided die

Let X be the number appearing on the die.



Each outcome from 1 to 6 is equally likely.

Outcome (x)	PMF: $P(X = x)$
1	$1/6$
2	$1/6$
3	$1/6$
4	$1/6$
5	$1/6$
6	$1/6$
Total	**1**

The PMF is:

$$P(X = x) = \begin{cases} 1/6 & ; x \in \{1, 2, 3, \dots\} \\ 0 & ; \text{other} \end{cases}$$

R.V.

→ PMF must satisfy two conditions:

① Pro. assign to each value must be $0 \le P \le 1$.

② Sum of the all pro. = 1.

→ PMF is function can be draw a graph of it ?

YES

Ex: Toss two coins

HH
HT
TH
TT

R.V. : No. of heads in two coin tosses

0 → 1
1 → 2
2 → 1

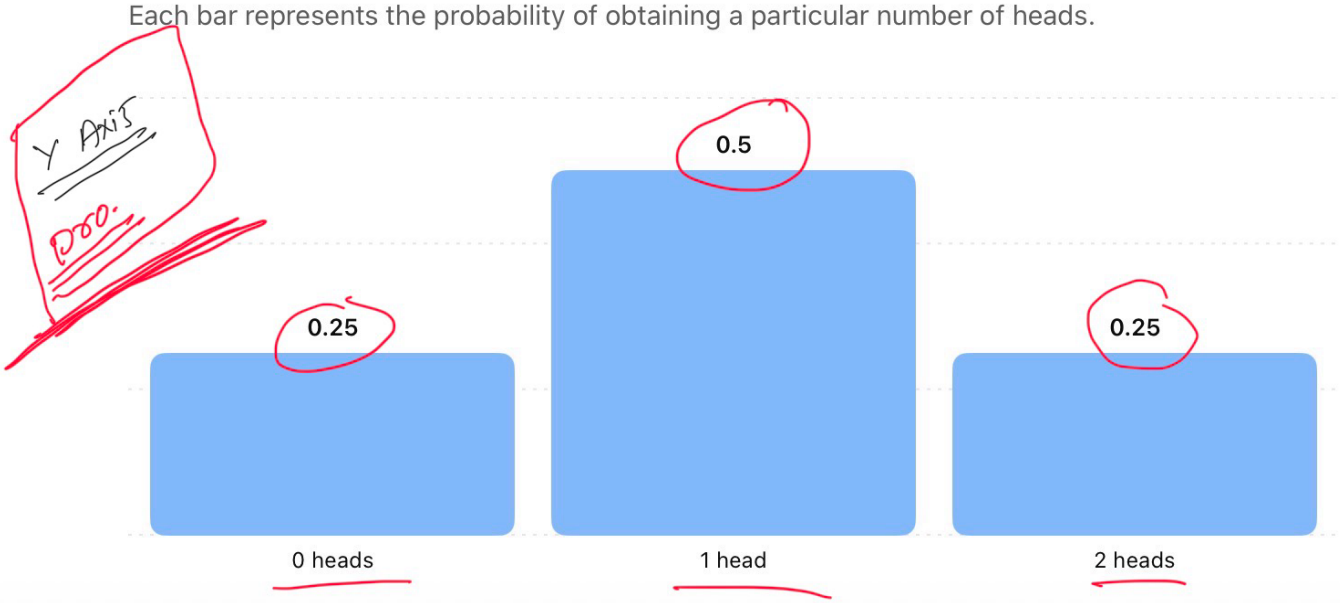
R.V.

PMF:

$$P(X=x) = \begin{cases} 1/4 & ; x = 0 \\ 2/4 & ; x = 1 \\ 1/4 & ; x = 2 \\ 0 & \text{other} \end{cases}$$

PMF: Number of heads in two coin tosses

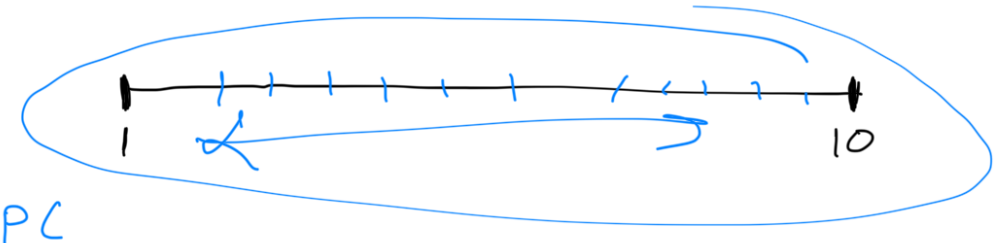
Each bar represents the probability of obtaining a particular number of heads.



→ What y Axis of PMF Rep. ?

- Prob. (0 ≤ val ≤ 1)

Q:



PC

→ We r 0 ~ 1 to 10 R..

$P(5) = (?) \rightarrow 0$

$= \frac{n(f)}{n(\infty)} = \frac{1}{\infty} = 0$

100 → 1000 → 10000



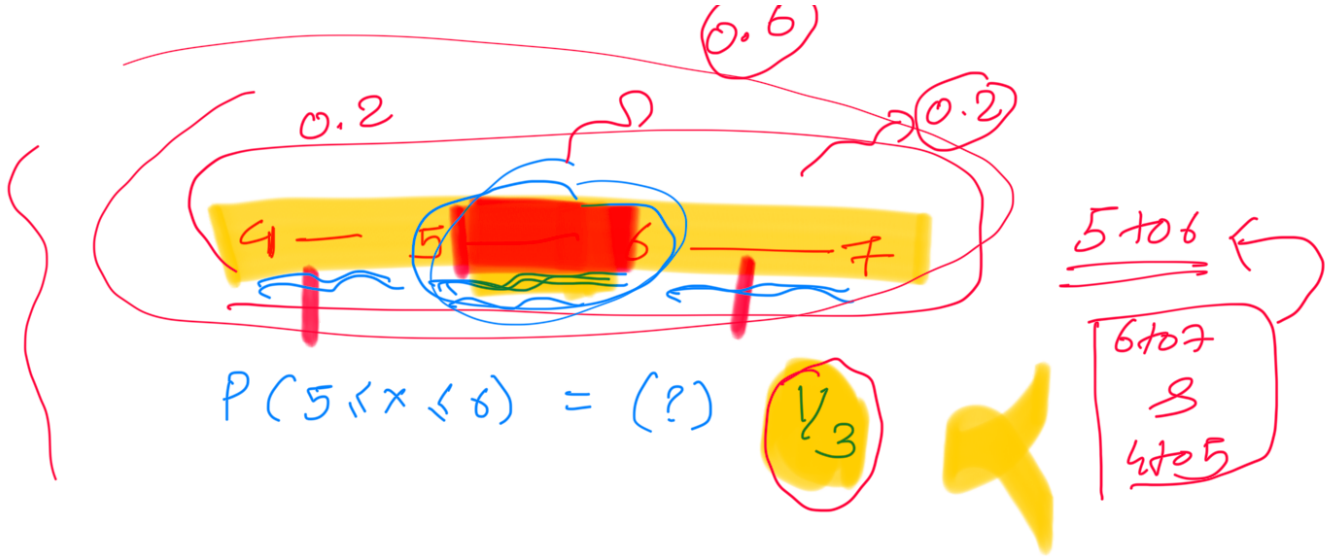
10 equal → Range

$\frac{1}{10}$ $\frac{4-3}{10-0}$

$P(0 \leq x \leq 10) = 1$

(2) GL Bajaj stu help/A

$\frac{0.6}{1}$

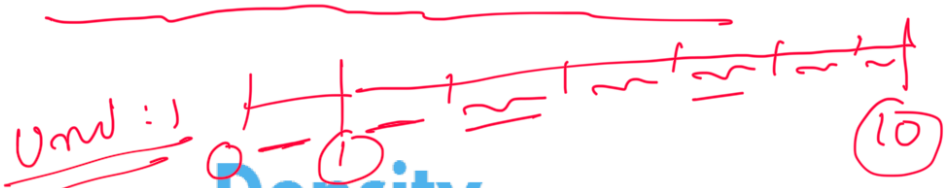


→ bcz pro. is not uniformaly distributed in this case.

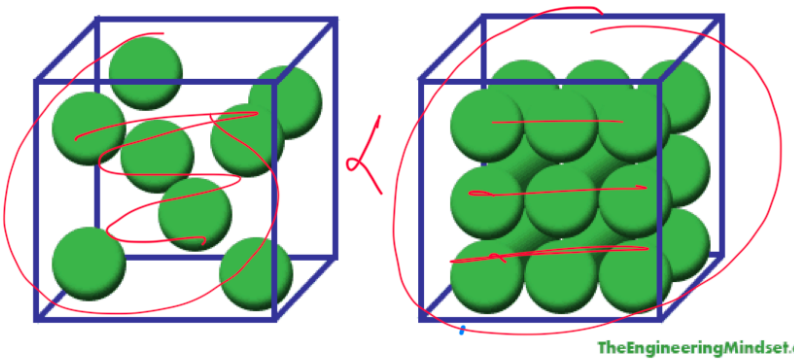
btwn
5 to 6 : higher number of students compare to 4 to 5 & 6 to 7.

Now what ?

→ Density ? mass per unit volume :



5 to 6
6 to 7
&
4 to 5



5 to 6 : higher no. of stud. → higher mass → high density

4 to 5 & 8 : Less density.
6 to 7

density: mass per Unit Volume:
length

pro.

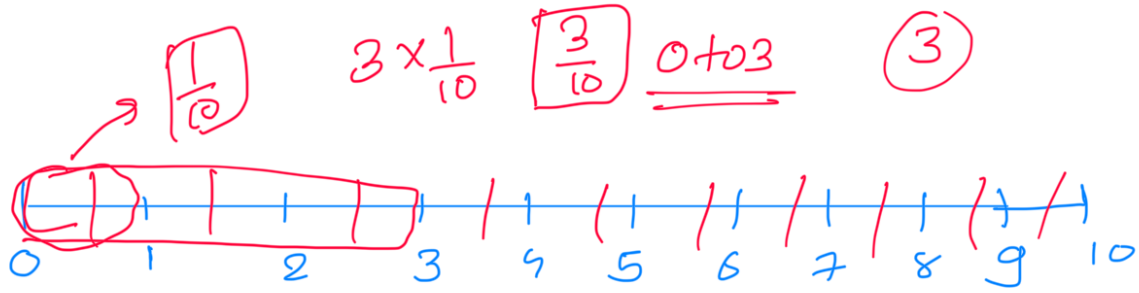
$\frac{m}{v}$

→ pro. per Unit length.

$$\text{density} = \frac{\text{total pro.}}{\text{total length}}$$

→ In our Example:

$3 \times \frac{1}{10}$ $\frac{3}{10}$ 0 to 03 (3)



$\frac{1}{10}$

total length = 10

total pro = 1

density = 1/10 → mass per Unit or pro. per Unit length.
(Unit means length = 1)

$$\text{density} = \frac{\text{total pro.}}{\text{total length}}$$

→ Now what is the pro. of having no. in 2 Unit length

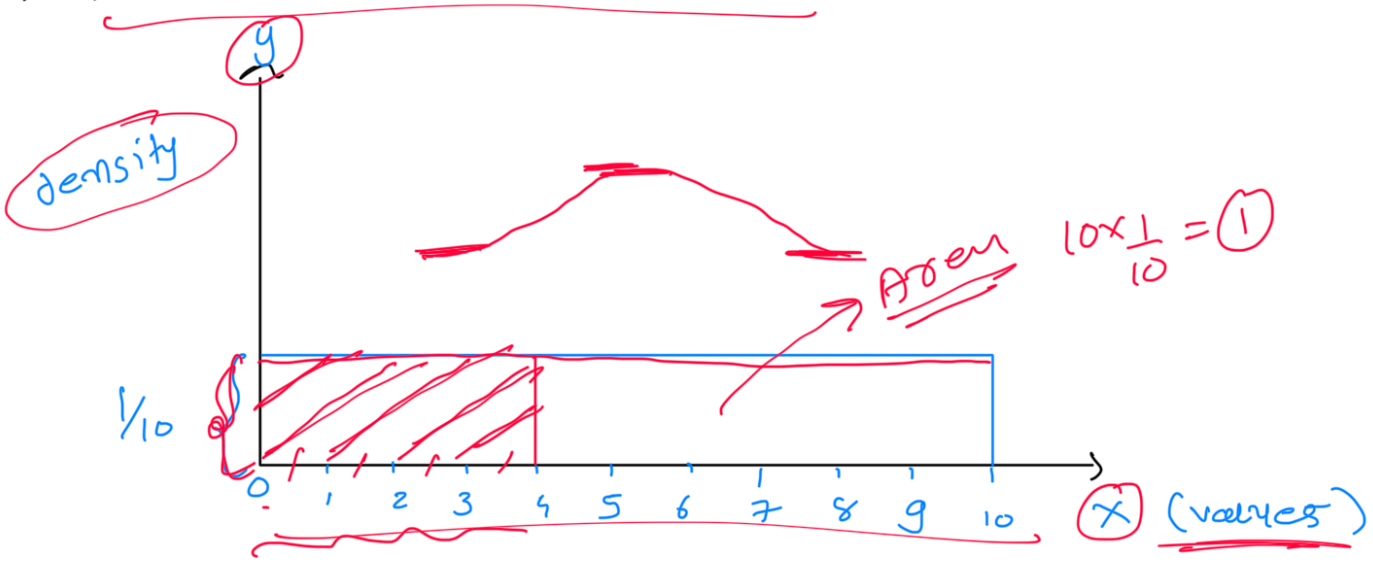
$= 2 \text{ (length)} * \frac{1}{10} \text{ (pro. per unit len.)}$

$= 2/10$

→ what is the pro. btwn 4 to 7 :

→ $3 * \frac{1}{10} = \frac{3}{10}$

→ now let's draw a graph :



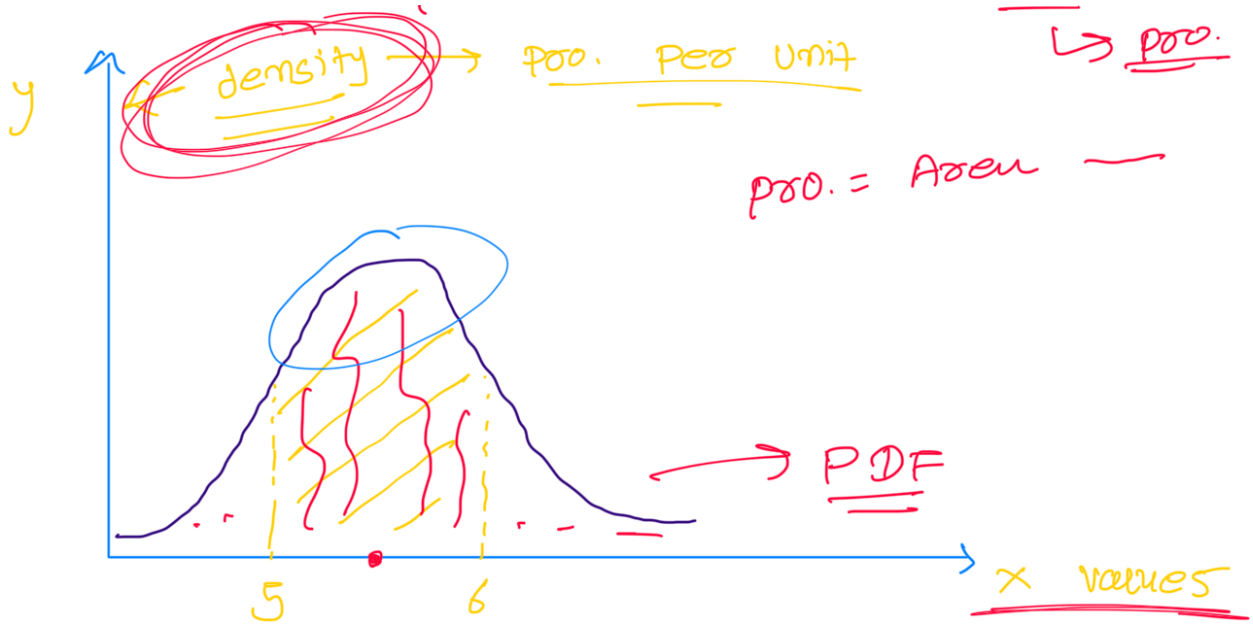
Area of Rectangle = hight × width

What is the pro. of number will be $\geq 4 \leq 7$

? = $\frac{4}{10}$ Area under the curve

→ now in the case of hight we have higher density btwn 5 to 6 :

PDF // PMF



pro $(5 < x < 6)$ = area under the curve

$$= \int_5^6 f(x)\, dx$$

* PDF (pro. density Function) :

- Describe the pro. of continuous R.V.
- total area under the curve = 1
- y Axis: density
- Why pro. density & why not pro. ?
- How to calc pro. = Area under the curve (Integration)

-> But how can we estimate the density ?

P(X <= 5)

③ CDF (cumulative distribution function): 0 to 5

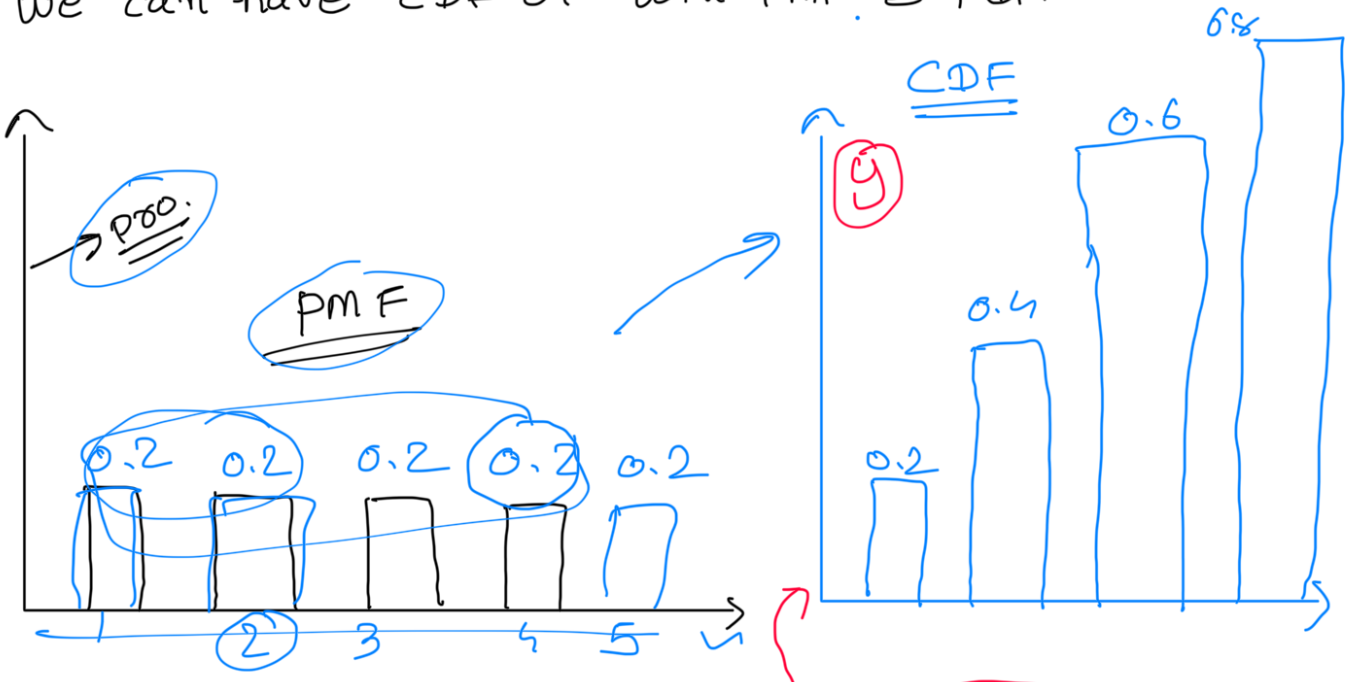
P(X <= x)

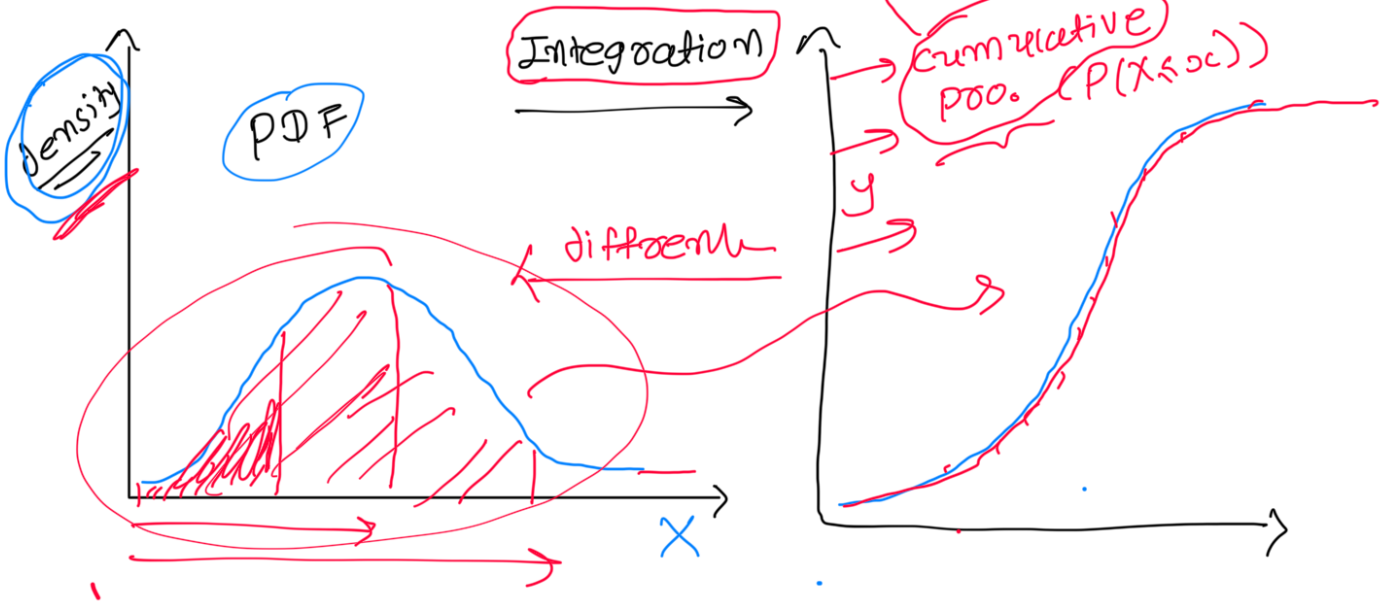
0 → x

0 10

Prob. that Random variable X will take a value less than or equal to x.

→ we can have CDF of both PMF & PDF.





* What is pending in Statistics ?

- Normal distribution
- Uniform / bernoulli / binom
- log normal
- CLT

Hypothesis